

Table 2: 10-shot segmentation accuracy on the individual tasks of the custom test set using the network trained with the best configuration.

Tasks	mIoU	mIoU (with outl. elim.)
Cup handle	0.820	0.820
Drill Machine Grip	0.838	0.838
Fork Grip	0.799	0.844
Screwdriver Grip	0.817	0.884
Tennis Racket Grip	0.585	0.644
<i>Average (all tasks)</i>	0.772	0.806

Table 3: Comparison of segmentation accuracy for 1-shot, 5-shot and 10-shot adaptation on the custom set tasks, using the network trained with the best configuration.

Tasks	mIoU		
	1-Shot	5-Shot	10-Shot
Cup Handle	0.218	0.557	0.820
Drill Machine Grip	0.275	0.634	0.838
Fork Grip	0.122	0.886	0.844
Screwdriver Grip	0.411	0.866	0.884
Tennis Racket Grip	0.127	0.55	0.644
<i>Average (all tasks)</i>	0.230	0.698	0.806

GAS test set and the custom test set, for 8-shot and 10-shot learning, respectively. We use different training loss functions and analyze the results. The network trained with data augmentation and using the sum of the BCE and Dice losses performed the best. The results for the individual tasks of the custom test set are given in Table 2, for the network trained with BCE+Dice loss and data augmentation. The qualitative results for the custom test set are shown in Figure 4.

Overall, we observe very good performance of grasp area prediction after adaptation using few samples. We also observed that the higher the variety in the few-shot adaptation set, the higher is the generalization performance of the networks. For instance, if the few-shot adaptation set includes only a single pose of an object, then the segmentation performance is lower on a test image that has the object at a different pose. A few-shot adaptation set with a lot of variety is shown in Figure 3.

We further analyze the performance of the network using different numbers of few-shot samples. Table 3 shows the outcome for the tasks in the custom test set. Even though we did not train our network for 1-shot learning, the mIoU values for 1-shot prediction are positive, showing that the estimated masks are partially overlapping with the correct grasp area. We observe that after 5-shot learning, the network already gives a reasonable output, which improves further for 10-shot learning. After 10-shot adaptation, the predicted masks are generally very close to the ground truth masks.

Finally, we demonstrate our framework on a robotic setup with a 7-DOF Franka Emika Panda manipulator, as shown in Figure 1. Here, the task is to grasp the handle of the

screwdriver. We use a state-of-the-art 6-DOF grasp estimator (?) that can output grasps within a given segmentation area. We use ROS MoveIt! framework (?) for collision-aware motion planning and execution. We observe a successful task-oriented operation based on grasp areas that are learned from few non-expert demonstrations, showing the effectiveness of our method.

Conclusion

In this paper, we proposed a task-oriented grasp area learning framework based on remote non-expert demonstrations. We use interactive segmentation to receive simple demonstrations from non-experts intuitively. We leveraged a meta learning strategy to prepare the segmentation network for few-shot adaptation and segment the task-oriented grasp regions. We combine the grasp area segmentation with a state-of-the-art grasp planner to plan and execute the 6-DOF grasp only on the allowed grasp area. For training our network, we created the GAS dataset with 10089 RGB images and grasp area segmentation masks, using a large-scale grasp dataset. Experimental evaluation showed the success of our method for task-oriented grasp area learning from few remote non-expert demonstrations. Future work can focus on improving the network architecture and exploring different meta learning algorithms for this task.

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